Index of KeyLogger

1. Definition of keylogger:

A keylogger is something that secretly watches what you type on your keyboard. It can be a hidden program or a physical device. When you type things like passwords, messages, or usernames, it records all of it and sends it to a hacker. The hacker can then use that information to break into your accounts.

2. What is keylogger?

A keylogger:

Runs secretly in the background. Records keystrokes (what you type). Sends that data to an attacker.

Example:

User types: password123

Keylogger records it and sends it → attacker now has login access

3. How Do keylogger work?

Keyloggers can operate at different levels of a computer's operating system. Some are software, meaning they are like hidden programs installed on your device. Others are hardware, meaning they are actual physical devices attached to your computer, usually requiring someone to access your device directly.

Modern keyloggers don’t just record what you type. They can also:

- Take screenshots
- Copy what you paste (clipboard)
- Record sound from your microphone
- Even access your webcam

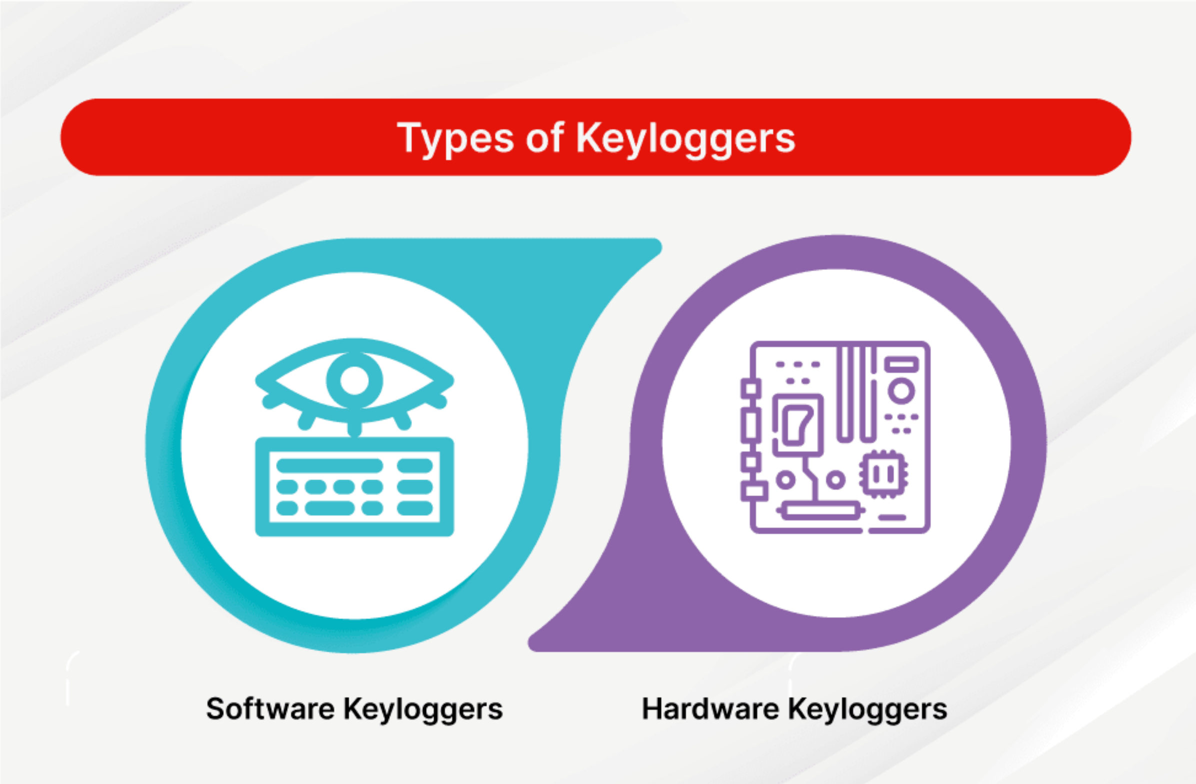
Their main goal is to steal private information like passwords, credit card details, and personal messages.

4. Types of Keyloggers:

There are two main types:

4.1 Software based Keylogger

4.2 Hardware based Keylogger



❖ **Software-Based Keyloggers:**

Software keyloggers are hidden programs that get installed on your computer, usually without you knowing. They can come from things like unsafe downloads, fake email attachments, or hacked websites.

Once inside your computer, they work like this:

- **Watching what you type:**

  They secretly "catch" your keystrokes before your computer or apps can use them and save everything you type.

- **Deep access (harder to detect):**

  Some advanced ones go deep into the system (the core part of your computer), which makes them harder to find and remove.

- **Sending your data away:**

  They regularly send the recorded information (like passwords or messages) to a hacker over the internet.

❖ **Hardware-Based keylogger:**

Hardware keyloggers are physical devices (actual objects) attached to a computer or keyboard. Someone usually has to physically access the device to install them.

They can be:

1. Plugged into a USB port
2. Placed between the keyboard and the computer cable
3. Built inside a keyboard or laptop
4. They work by recording what you type before it even reaches the computer, so the system doesn't notice anything unusual. Because they are physical devices (not software), they are often harder to detect.

5. Why are KeyLoggers Used?

You might think keyloggers are always illegal, but that's not true everywhere. They can be used for good reasons too.

For example:

- IT teams use them to fix computer problems
- Companies may use them to check if employees are misusing data
- Parents may use them to monitor their children's computer use

But keyloggers become dangerous and illegal when used with bad intent.

Simple rule:

- **Legal:** If you install it on your own device or with permission
- **Illegal:** If you secretly install it on someone else's device to steal information

❖ **Legitimate (legal) uses of keyloggers:**

Keyloggers can be used in a **safe and legal way** when they are used for protection, monitoring, or improving security.

- **Security testing:**
  Experts (ethical hackers) use them to find weaknesses in systems so they can fix them.
- **Parental control:**
  Parents use them to keep an eye on their children's online activity and keep them safe.
- **Employee monitoring:**
  Some companies use them to check work activity and make sure company resources are used properly.

- **Malicious (illegal) uses of keyloggers:**

Keyloggers become dangerous when they are used to **spy, steal information, or harm others**.

- **Cybercrime:**
  Hackers use them to steal passwords, credit card details, and personal data for fraud or identity theft.
- **Espionage:**
  They can be used to secretly collect important or confidential information from companies or governments.
- **Cyberbullying / spying:**
  Someone may use them to watch another person's private messages and activities, invading their privacy.
- **RATs (Remote Access Trojans):**
  Some malware includes keyloggers, letting attackers **spy on and control** a victim's computer from far away.

6. How keyloggers attack your device:

Keyloggers need to get into your device first. They do this in a few common ways:

- **Spear phishing:**

  You get a fake email or message that looks real. When you click the link or open the file, the keylogger gets installed.

- **Drive-by download:**

  You visit a bad website, and the keylogger installs automatically without you knowing.

- **Trojan horse:**

  You download something that looks safe (like a file or app), but it secretly contains a keylogger.

- **Problems caused by keyloggers (computers)**

- **Slower performance:**

  Your computer may become slow because the keylogger is running in the background.

- **Typing delay:**
  Letters may appear slowly when you type.

- **Apps freezing:**
  Programs may hang or crash more often.

- ❖ **On phones (Android & iPhone)**

Even phones can be affected by software keyloggers.

They can:

- Track what you tap on the screen
- Take screenshots
- Record audio or camera activity
- Monitor internet usage
- They may get installed if:
- Someone briefly accesses your phone
- You click a harmful link or download a bad app

7. Signs there might be a keylogger on your device:

- **Your device becomes slow**

  It lags or feels heavy even when doing simple tasks.

- **Strange programs running**
  You see unknown apps or processes you don't recognize.
- **Antivirus finds something**
  A scan shows malware or suspicious files.
- **Unknown apps installed**
  You notice software you never installed.
- **Unusual internet activity**
  Data is being sent somewhere even when you're not using the internet much.
- **Typing feels delayed**
  Letters appear slowly when you type.
- **Apps freeze or crash**
  Programs stop working randomly.
- **Weird device behavior**
  Pop-ups, glitches, or unusual changes happen.
- **Something attached to your computer**
  A strange USB or device is plugged in (possible hardware keylogger).

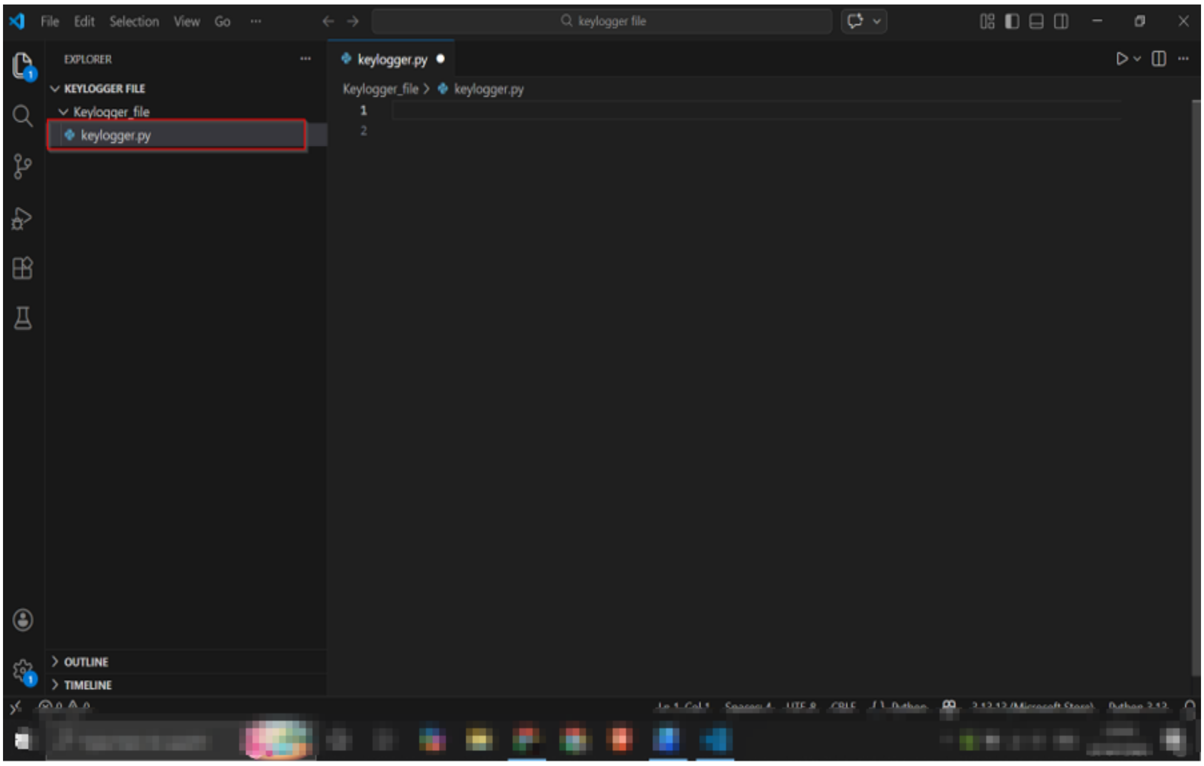
8. **To protect yourself from KeyLoggers:**

- **Use security software**
  Install antivirus to detect and block keyloggers.
- **Keep everything updated**
  Update your system and apps to fix security holes.
- **Be careful online**
  Don't click unknown links or download suspicious files.
- **Use a virtual keyboard**
  For passwords, you can use an on-screen keyboard to stay safer.
- **Scan your device regularly**
  Run antivirus checks to find and remove threats.
- **Check physical safety**
  Look for unknown devices attached to your computer, especially in public places.

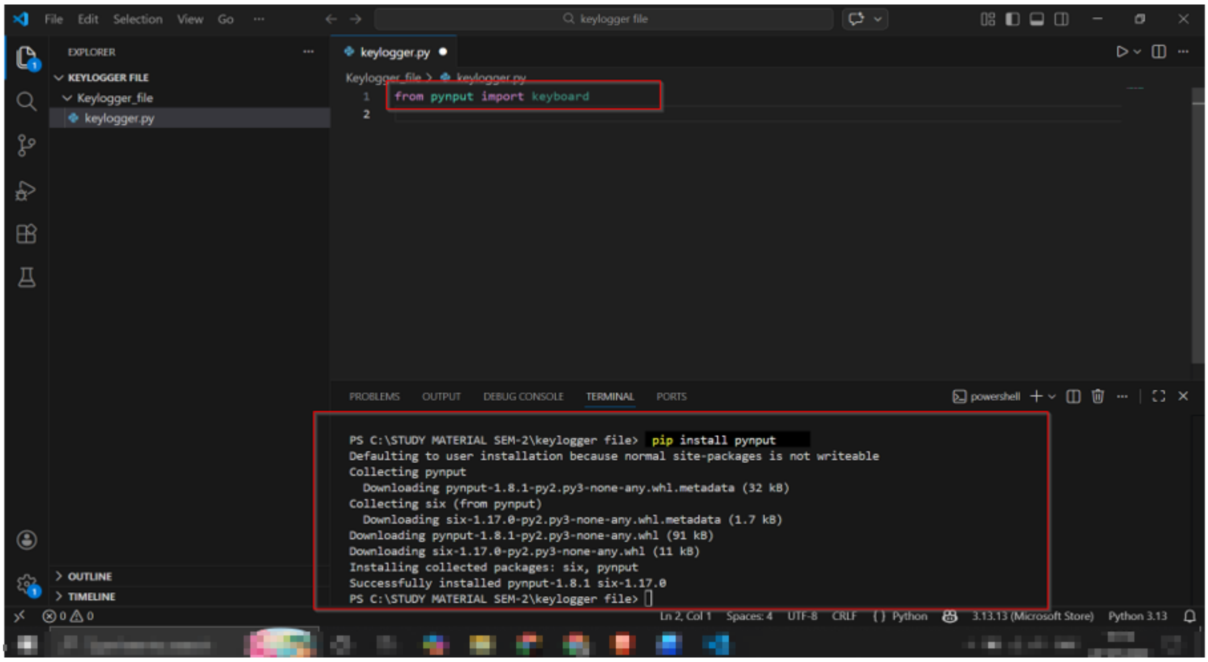
Practical Hand On

❖ **Create a KeyLoggers help of code using python.**

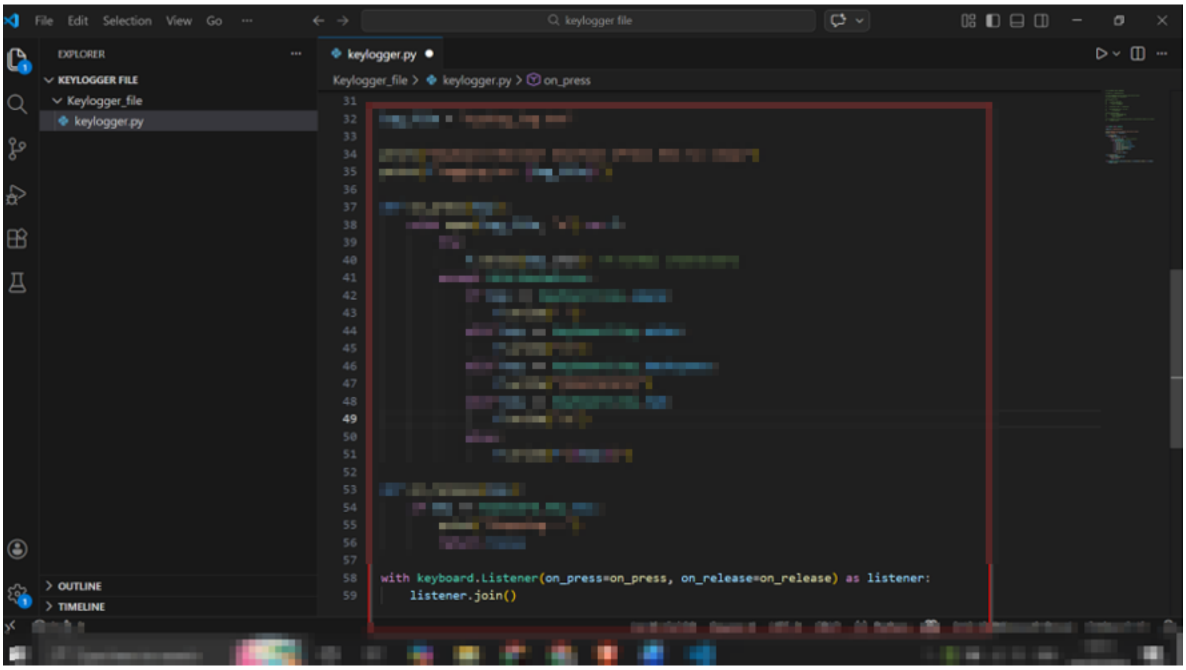
Step-1:- First, open Visual Studio Code and make sure Python is installed. Then create a new file and save it as **keylogger.py** to start writing your Python code.



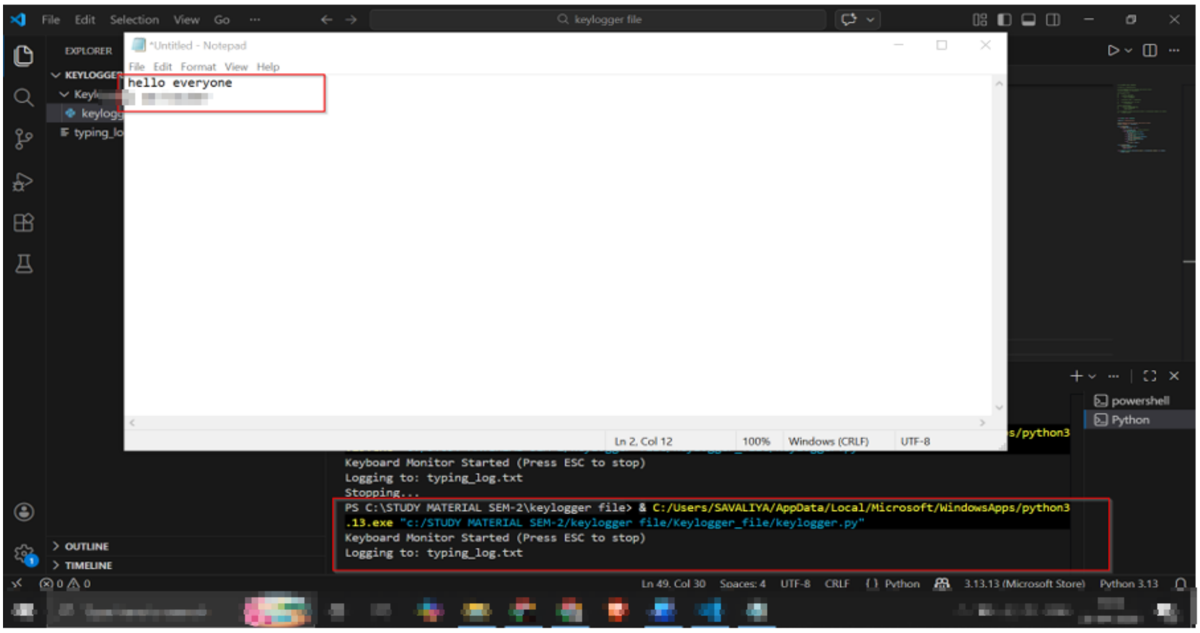
Step-2:- import the required library in your Python file using from pynput import keyboard. Then install the pynput package using the command pip install pynput in the terminal.



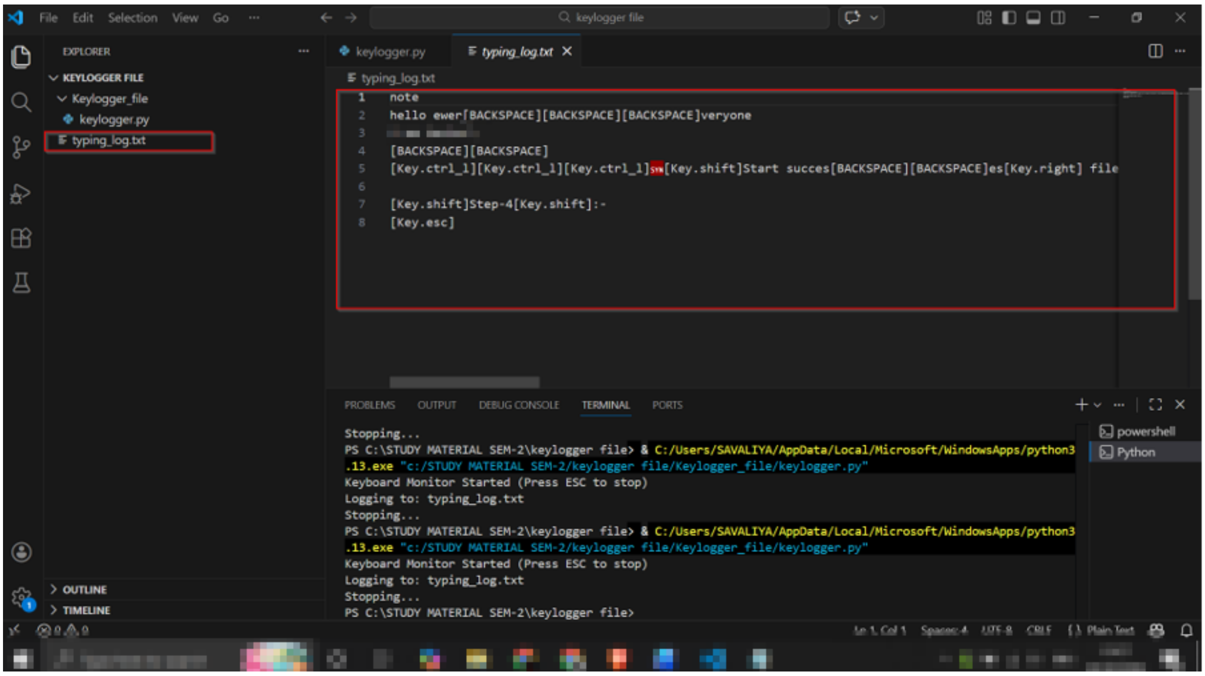
Step-3:- This Python program listens to your keyboard input and records every key you press. It saves the typed data into a file (typing_log.txt) while the program is running. The program stops when you press the **ESC key**. Run this file.



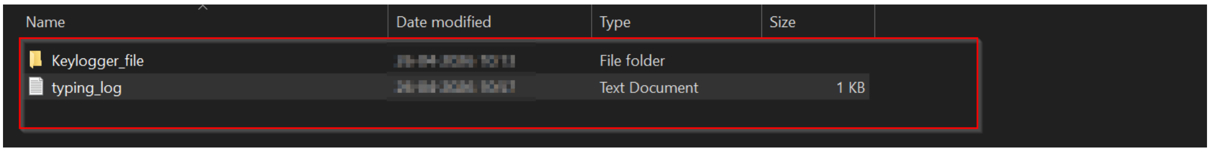
Step-4:- The program is running successfully in the background and logging keystrokes into a file (typing_log.txt). When you type “hello everyone” in Notepad, the same input is captured and saved by the Python script. This shows that the keyboard monitor is working and recording your typing in real time.

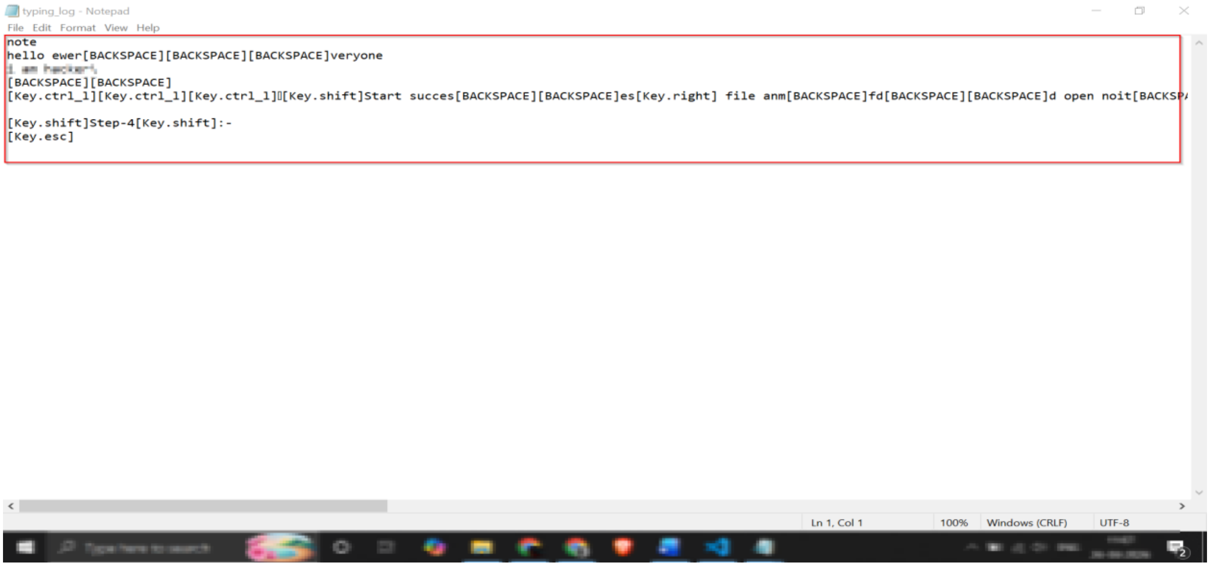


Step-5:- The program records all the keys you press and saves them into the typing_log.txt file. It logs both normal text and special keys (like backspace, shift, ctrl), which is why you see entries like [BACKSPACE] in the file. This shows that the script is capturing and storing your typing activity exactly as it happens.



Step-6:- The program creates a log file (typing_log.txt) in the main project folder and saves all the keys typed by the user. It records both normal text and special keys like backspace, shift, and enter. This demonstrates how keyboard input is captured and stored in real time.





Conclusion: This project helped me understand how keyloggers capture and store keyboard input in real time. It shows how both normal and special keys can be recorded using Python. This knowledge is useful for learning cybersecurity concepts and understanding potential risks. The project is done only for educational and ethical purposes.

References

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https://www.sophos.com/en-us/cybersecurity-explained/keylogger